

Prime Dynamics Program, Volume II

Physical Riesz Packets, Temporal Clouds, and the Trace-Envelope Frontier
A Provenance-Preserving Synthesis of RH-161–RH-241

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Abstract

This volume synthesizes RH-161–RH-241, the eighty-one-paper interval that turns the conditional Stage-A language of Volume I into a typed packet-to-Riesz, relative-determinant, temporal-cloud program. RH-161 proves an abstract fixed-contour assembly theorem. If a packet block has resolvent bound M , off-packet coupling ε , and $M\varepsilon < 1$, spectral rank is stable; the quantitative graph margin

$$\delta = \frac{|\Gamma|M^2\varepsilon}{2\pi(1-M\varepsilon)} < 1$$

gives an invertible packet-to-Riesz map. Complementary Schatten convergence, pole accounting, normalization, and directed marked traces then define a conditional enhanced datum for determinant types $p = 1$ and $p = 2$ without identifying the two branches.

The following eighty papers develop physical Riesz shells, history/cycle algebra, biorthogonal Feshbach reductions, source-channel quotients, transport obstructions, quartet gauges, rank-growing reciprocal clouds, and projection-free $\det_2$ factors. The endpoint RH-240 theorem is an exact criterion: a uniform moving all-order noisy trace envelope would yield a normal relative-determinant family. RH-241, however, certifies only orders 2–12 and leaves both the all-order envelope and coefficient anchor open.

We prove an executable provenance theorem, restate the typed assembly theorem with its physical hypotheses visible, and give a finite-jet nonpromotion theorem. The moving noisy all-order trace envelope remains open. Gates A–E remain false/open. This synthesis does not construct a Hilbert–Polya operator, identify Riemann zeros, prove a von Mangoldt trace, prove completed-zeta divisor equality, or prove the Riemann Hypothesis.

Keywords: Riesz projections; packet reductions; relative determinants; temporal traces; reciprocal clouds; trace envelopes; provenance audit.

1 Scope and source hierarchy

Volume II begins at RH-161 rather than RH-160 because RH-161 is the first paper to state the complete typed assembly interface. It is followed by eight ten-paper phases ending at RH-171, 181, 191, 201, 211, 221, 231, and 241. Review papers summarize status and finite counts; analytic claims remain sourced to the individual theorem papers.

We distinguish analytic theorems, finite certificates, conditional implications, scoped negatives, and open/not-testable leaves. A packet in an abstract finite algebra is not automatically an actual noisy spectral packet. A local probability law is not a trace observation without an identification theorem. A finite list of moments is not an all-order analytic envelope.

2 Executable provenance theorem

Theorem 2.1 (Volume II provenance theorem). *The executable inventory discovers exactly one complete source directory for each integer $161 \leq n \leq 241$. It partitions the eighty-one labels into*

the singleton RH-161 assembly phase and eight consecutive ten-paper phases, records the nine anchors, and binds selected source files by SHA-256.

Proof. Directory discovery uses the numerical prefix. A duplicate or a source missing its README, TeX manuscript, or PDF is rejected. The integer list and the union of the phase intervals are checked against the full source range. The builder hashes available theorem ledgers, roadmaps, and result records; the archive verifier independently replays all local and external hashes. This proves source identity rather than re-proving eighty-one papers. $\square$

The eight review artifacts contain 30,435 explicitly defined finite items in total. Their per-review counts are stored in the machine audit. These counts document breadth and internal consistency only.

3 RH-161: the typed assembly theorem

Let a finite packet block be isolated by a contour Γ . Write M for the declared block resolvent supremum and ε for the off-packet coupling. RH-161 proves rank stability under

$$M\varepsilon < 1$$

and an invertible graph bridge under the stronger bound

$$\delta = \frac{|\Gamma|M^2\varepsilon}{2\pi(1 - M\varepsilon)} < 1. \quad (1)$$

The assembly then requires one of two reset seeds together with five physical interfaces:

$$(S_{\text{native}} \vee S_{\text{lagged}}) \wedge R \wedge Q \wedge U \wedge Z \wedge T. \quad (2)$$

Here R is the physical packet-to-Riesz lift, Q the cloud coefficient and pole ledger, U a common-space Schatten complement limit, Z canonicity and normalization, and T directed marked-trace convergence.

Theorem 3.1 (Typed assembly theorem). *Under one seed and all five interfaces in (2), exact moving cloud division, complementary $\mathcal{S}_p$ convergence, complete pole accounting, target-independent normalization, schedule independence, and a fixed separating family of directed marked traces produce an enhanced relative-determinant datum. The theorem covers $p = 1$ and $p = 2$ as typed alternatives and does not identify them.*

Proof. The packet homotopy and (1) identify a same-rank Riesz space. Exact reducing factorization divides out the finite cloud. Common-space $\mathcal{S}_p$ convergence gives locally uniform $\det_p$ convergence for the complement. Q restores the deterministic pole divisor, Z fixes the zero-free factor and removes schedule/gauge dependence, and T retains the directed finite words. Every step uses a named hypothesis. None of R, Q, U, Z, T is proved for the all-level prime-dynamics family by RH-161. $\square$

The two inclusion-minimal completion bundles are therefore

$$\{S_{\text{native}}, R, Q, U, Z, T\}, \quad \{S_{\text{lagged}}, R, Q, U, Z, T\}.$$

This is an exact dependency statement, not Gate-A closure.

4 RH-162–RH-171: physical Riesz shells

This block translates the assembly variables into shellwise physical obligations. It distinguishes rank isolation, graph stability, coefficient transport, complement control, and temporal marking. The RH-171 review contains 3,584 finite cases and 63 omission witnesses. Its four-leaf frontier is a conditional implication: satisfying the named physical leaves would yield shell conclusions.

The finite cases test formulas and logical minimality. They do not provide a uniform contour or a small-noise coupling theorem. A globally bounded projection norm is neither proved nor required as a substitute for the typed local graph estimates.

5 RH-172–RH-181: history and cycle realization

The history/cycle phase proves exact algebra connecting finite temporal histories, cyclic closures, markers, and determinant contributions. It identifies which finite words retain orientation and which quotient operations erase it. The 2,600-item review verifies the declared finite identities.

A literal infinite-history Fredholm construction is stopped in its named form: the needed compactness and uniform trace-class structure are absent. The negative conclusion does not reject every adapted history space. The surviving route uses finite separating words transported through an actual common-space limit.

6 RH-182–RH-191: biorthogonal and Feshbach structure

These papers develop balanced left/right frames, exact Feshbach identities, and Schur-budget criteria for nonnormal packets. The point is not to make the operator normal, but to control the correct biorthogonal coordinates and resolvent factors. The RH-191 audit contains 2,960 finite items.

The conditional Riesz conclusions still require validated physical block separation and coupling. Orthogonal finite clocks and unqualified condition numbers cannot replace those inputs. A lower bound on conditioning is not an upper stability theorem.

7 RH-192–RH-201: source-channel quotients

This phase distinguishes source-cyclic from source-observation channels and constructs exact quotient and packet dictionaries. It proves canonicity relative to the declared finite source algebra and audits 1,352 items.

The word “canonical” remains typed. It does not prove independence from every mesh, schedule, reset, or cutoff in the physical continuum family. Those are the Z obligations of (2). Nor does a source channel by itself supply the cloud coefficient bridge Q .

8 RH-202–RH-211: transport failures and divisor-first pivot

Exact intertwining and transport budgets expose the cost of moving quartets and edge clouds between representations. Several natural routes fail: naive Haar transport, scalar residue renormalization, and an expanding-edge cloud transport lack the required uniformity or normalization. RH-210 and RH-211 therefore pivot to a divisor-first formulation. The review contains 649 finite records.

This pivot is a change in proof organization, not a zeta-divisor theorem. The physical divisor, coefficient bridge, and complete no-missing/no-spurious statement remain absent.

9 RH-212–RH-221: quartet shapes and fixed degree

Centered conjugate quartets admit intrinsic shape coordinates, monotone axial clocks, boundary strata, and gauge-complete reconstruction formulas. The phase also proves transverse quenching and recurrence obstructions. These are exact results in the quartet data type, supported by a 2,140-item review.

A fixed quartic has fixed degree and cannot by itself encode a rank-growing spectral cloud. This is a structural obstruction to a fixed-quartic route, not to all rank-growing packet constructions. Gauge-complete reconstruction of one divisor shape does not identify a physical all-level divisor.

10 RH-222–RH-231: reciprocal clouds and determinants

This block builds rank-growing conjugate cloud atlases, reciprocal dictionaries, empirical tightness criteria, and regularized determinant formulas. Projection-free descriptions reduce dependence on unstable projector coordinates. The RH-231 review audits 9,870 finite items.

The determinant formulas are exact for their declared clouds. They do not show that the cloud is the actual physical resonance divisor, that its rank matches a canonical spectral count, or that its complement has a uniform common-space ideal limit. Direct promotion from resonance samples to a completed divisor is therefore stopped.

11 RH-232–RH-241: the trace-envelope wall

The final phase constructs projection-free finite $\det_2$ factors and analyzes which coefficient information controls a normal analytic family. Finite jets can match while higher coefficients vary arbitrarily; local polynomial data therefore do not imply determinant convergence.

RH-240 states the positive all-order criterion. Schematically, a uniform geometric envelope for every moving noisy trace coefficient, together with the correct coefficient identification and analytic tail, yields local normality and a relative-determinant limit. RH-241 audits 7,280 finite items and obtains a subunit envelope only for orders 2–12.

Theorem 11.1 (Finite-jet nonpromotion theorem). *Control of a fixed finite set of Taylor or trace coefficients, even with arbitrarily accurate finite fits, does not imply a uniform all-order analytic envelope or convergence of the associated determinant family.*

Proof. For any fixed jet order m and nonzero z_0 , the zero-free entire functions

$$F_K(z) = \exp\left(K(z/z_0)^{m+1}\right)$$

have identical Taylor coefficients through order m while their values and higher coefficients are unbounded as K varies. Thus finite jets do not control a normal family. The same logical gap applies to the orders 2–12 ledger unless a uniform all-order source theorem is added. $\square$

The endpoint has two separate missing leaves:

- (i) the moving noisy all-order trace envelope; and
- (ii) a no-over-extraction coefficient anchor identifying the selected noisy cloud coefficients with the determinant numerator.

The deterministic coefficient dictionary developed later in Volume III solves a target-side problem and does not retroactively prove either leaf.

12 Frontier and reopening conditions

The physical route may reopen through an actual moving contour/coupling theorem, a common-space $\mathcal{S}_1$ or $\mathcal{S}_2$ complement limit, complete $Q/Z/T$ identifications, or a direct all-order noisy envelope with coefficient bridge. Each input must use one clock, normalization, and operator data type.

Fixed finite endpoint extensions, synthetic clouds, local probability models, and finite regression do not reopen the all-order leaf. The route coordinate is

physical_riesz_cloud_assembled_moving_noisy_trace_envelope_open.

13 Reproduction and claim boundary

The volume audit verifies the source interval, nine anchors, nine phase ranges, finite review counts, determinant-type adapter, open physical interfaces, fixed-order boundary, Gate values, and forbidden macro claims. It binds selected upstream files and all local publication artifacts by SHA-256. The tests enforce that the fixed-order maximum remains 12 and that neither the all-order envelope nor coefficient anchor is promoted.

The audit verifies identity and declared scope rather than re-proving the source corpus. The moving noisy all-order trace envelope remains open. Gates A–E remain false/open. Volume II does not construct a Hilbert–Polya operator, identify Riemann zeros, prove a von Mangoldt trace, prove equality with the completed-zeta divisor, or prove the Riemann Hypothesis.

A Atomic source index

This machine-generated table exposes every canonical numbered source in the volume. It is a provenance index; theorem status remains in the source ledger and the phase crosswalk.

source	canonical directory
RH-161	RH-161-packet-riesz-relative-determinant-assembly
RH-162	RH-162-ambient-realization-reset-riesz-gate
RH-163	RH-163-two-sided-schur-packet-riesz-certificate
RH-164	RH-164-balanced-similarity-packet-coupling
RH-165	RH-165-midgap-normal-block-contour
RH-166	RH-166-bi-ritz-directional-riesz-graph
RH-167	RH-167-finite-mesh-resolvent-envelope
RH-168	RH-168-operator-ball-mesh-schur-transfer
RH-169	RH-169-common-coordinate-riesz-transport
RH-170	RH-170-rank-growing-riesz-shell-atlas
RH-171	RH-171-ten-layer-physical-riesz-interface-review
RH-172	RH-172-canonical-polar-reset-memory-realization
RH-173	RH-173-normalized-history-cocycle
RH-174	RH-174-physical-history-realization-audit
RH-175	RH-175-infinite-history-shift-obstruction
RH-176	RH-176-finite-cyclic-history-closure
RH-177	RH-177-double-cycle-geometric-cloud
RH-178	RH-178-orientation-marked-cycle-traces
RH-179	RH-179-clock-cycle-cloud-degree-calibration
RH-180	RH-180-cyclic-cloud-riesz-shell-theorem
RH-181	RH-181-ten-layer-history-cycle-route-review

source	canonical directory
RH-182	RH-182-finite-temporal-clock-physical-audit
RH-183	RH-183-projective-wrap-obstruction
RH-184	RH-184-balanced-biorthogonal-temporal-realization
RH-185	RH-185-physical-bi-krylov-cycle-calibration
RH-186	RH-186-oblique-conditioning-riesz-budget
RH-187	RH-187-regularized-cross-gram-tradeoff
RH-188	RH-188-directional-coupling-gauge-balance
RH-189	RH-189-oblique-feshbach-determinant-factorization
RH-190	RH-190-complement-resolvent-budget-audit
RH-191	RH-191-ten-layer-biorthogonal-frontier-review
RH-192	RH-192-frobenius-left-multiplication-obstruction
RH-193	RH-193-source-cyclic-spectral-quotient
RH-194	RH-194-physical-edge-root-matching
RH-195	RH-195-source-observation-riesz-channels
RH-196	RH-196-canonical-biorthogonal-spectral-packet
RH-197	RH-197-physical-residue-transversality-audit
RH-198	RH-198-temporal-spectral-packet-alignment
RH-199	RH-199-source-channel-determinant-trace-factorization
RH-200	RH-200-conjugate-pair-edge-quartet-selection
RH-201	RH-201-ten-layer-source-channel-review
RH-202	RH-202-adjacent-edge-quartet-transport
RH-203	RH-203-riesz-intertwining-transport-budget
RH-204	RH-204-conjugate-branch-correspondence
RH-205	RH-205-endpoint-procrustes-shell-map
RH-206	RH-206-residue-cocycle-renormalization-obstruction
RH-207	RH-207-dual-channel-quartic-divisor-flow
RH-208	RH-208-endpoint-isolation-transport-certification
RH-209	RH-209-expanding-edge-cloud-transport-obstruction
RH-210	RH-210-divisor-first-gate-a-pivot
RH-211	RH-211-ten-layer-transport-frontier-review
RH-212	RH-212-intrinsic-quartic-normalization-audit
RH-213	RH-213-centered-conjugate-quartet-shape-manifold
RH-214	RH-214-monotone-axial-shape-clock
RH-215	RH-215-out-of-sample-shape-clock-prediction
RH-216	RH-216-degenerate-quartet-boundary-stratification
RH-217	RH-217-transverse-shape-quenching
RH-218	RH-218-autonomous-shape-recurrence-obstruction
RH-219	RH-219-fixed-quartic-counting-obstruction
RH-220	RH-220-gauge-complete-divisor-reconstruction
RH-221	RH-221-ten-layer-quartet-shape-frontier-review
RH-222	RH-222-rank-growing-conjugate-cloud-atlas
RH-223	RH-223-shell-complete-edge-selection-stability
RH-224	RH-224-global-cloud-gauge-tightness
RH-225	RH-225-tight-cloud-local-finiteness-obstruction
RH-226	RH-226-reciprocal-resonance-fredholm-dictionary
RH-227	RH-227-reciprocal-local-count-small-noise-gate
RH-228	RH-228-resolved-det2-omitted-shell-control
RH-229	RH-229-nonnormal-frobenius-tail-budget-barrier
RH-230	RH-230-dual-channel-det2-coherence-noncontraction
RH-231	RH-231-ten-layer-rank-growing-determinant-frontier-review

source	canonical directory
RH-232	RH-232-biorthogonal-riesz-cloud-projection
RH-233	RH-233-radial-gap-pseudospectral-barrier
RH-234	RH-234-projection-free-det2-spectral-factor
RH-235	RH-235-trace-vs-hilbert-schmidt-separation
RH-236	RH-236-cloud-extracted-trace-moment-atlas
RH-237	RH-237-dual-channel-trace-jet-coherence
RH-238	RH-238-trace-adaptive-shell-selection
RH-239	RH-239-adaptive-jet-contraction-obstruction
RH-240	RH-240-uniform-trace-envelope-criterion
RH-241	RH-241-ten-layer-trace-envelope-frontier-review